

YUN-RONG (LAUREN) LUO

✉ yunrong@umich.edu | 🌐 <https://lauren-yrlo.github.io> | 🎓 Google Scholar

RESEARCH INTERESTS

Distributed systems, formal verification, system performance, and programming languages

EDUCATION

University of Michigan, Ann Arbor (UMich)

Ph.D. Candidate in Computer Science and Engineering

Ann Arbor, MI

08/2023-2029

- Research Advisor: Prof. Manos Kapritsos
- Coursework: Distributed Systems, Formal Verification of Systems Software, Advanced Programming Languages, AI Foundations

National Taiwan University (NTU)

M.S. in Electronics Engineering

Taipei, Taiwan

09/2021-2023

- Research Advisor: Prof. Jie-Hong Roland Jiang
- Coursework: Advanced Algorithms, Graph Theory, Programming Languages, Physical Design for Nanometer Ics

B.S. in Electrical Engineering

09/2017-06/2021

- Coursework: Data Structures, Algorithms, Operating Systems, Embedded Systems Lab, Logic Synthesis and Verification, System-on-Chip Verification, Electronic Design Automation, Computing Theory, Complexity Theory, Mathematical Logic

PUBLICATIONS

Conference

1. **Yun-Rong Luo**, Aman Goel, Karem Sakallah, "QSM-Cutoff: Systematic Derivation of Quantified Cutoff Formulas for Distributed Protocols." 2025 International Conference on Computer Aided Verification (CAV), 2025 [[link](#)]
2. Che Cheng*, **Yun-Rong Luo***, Jie-Hong Roland Jiang, "Knowledge Compilation for Incremental and Checkable Stochastic Boolean Satisfiability." 2024 International Joint Conference on Artificial Intelligence (IJCAI), 2024 [[link](#)]
3. Yu-Neng Wang*, **Yun-Rong Luo***, Po-Chun Chien*, Ping-Lun Wang, Hao-Ren Wang, Wan-Hsuan Lin, Jie-Hong Roland Jiang, Chung-Yang Ric Huang, "Compatible Equivalence Checking of X-Valued Circuits." 2021 IEEE/ACM International Conference on Computer-Aided Design (ICCAD), 2021 [[link](#)]

Journal

1. **Yun-Rong Luo***, Che Cheng*, Jie-Hong Roland Jiang, "A Resolution Proof System for Dependency Stochastic Boolean Satisfiability." Journal of Automated Reasoning, 2023 [[link](#)]

RESEARCH EXPERIENCE

Graduate Student Research Assistant (UMich)

Performance Verification for Distributed Systems

Ann Arbor, MI

2025-Present

- Developing formal performance guarantees for distributed systems, including latency and throughput bounds.
- Applying stochastic network calculus to derive performance bounds for distributed systems.

Formal Verification for Distributed Protocols

2023-2025

- Developed [QSM-Cutoff](#): an automated inductive invariant inference tool for distributed protocols.
- Enhanced forward-reachability and inductive-invariant inference algorithms using symmetry and logic minimization.

Graduate Student Research Assistant (NTU)

Certified Stochastic Boolean Satisfiability Solving

Taipei, Taiwan

2023-2024

- Enabled proof generation for the stochastic Boolean satisfiability (SSAT) solver [SharpSSAT](#) via knowledge compilation.
- Extended a certified model-counting toolchain into [cert-SSAT](#) for certifying SharpSSAT results and helped identify a bug.

Proof System for Dependency Stochastic Boolean Satisfiability

2021-2023

- Developed the first sound and complete resolution proof system for dependency stochastic Boolean satisfiability (DSSAT).
- Formalized the DSSAT proof system and proved its soundness and completeness.

Undergraduate Student Researcher (NTU)

Equivalence Checking of X-Valued Circuits

Taipei, Taiwan

2020-2021

- Contributed to the first-place solution for *equivalence checking of X-valued circuits* at the 2020 ICCAD CAD Contest.
- Investigated SAT encodings for verification and optimized the circuit rewriting algorithm on 28 industrial cases.

WORK EXPERIENCE

Cadence Design Systems

R&D Intern, Formal Verification Team

Hsinchu, Taiwan

07/2020-09/2020

- Generated *engineering change order (ECO)* test cases for the Conformal ECO tool.

SELECTED PROJECTS

EECS 491: Introduction to Distributed Systems (UMich)

Winter 2026

- Implemented the following systems in Go:
 - a primary/backup key-value service, a Paxos-based key-value service, and a sharded key-value service with Paxos groups.

CSE 498/598: Formal Verification of Systems Software (UMich)

Fall 2025

- Verified a sharded key-value store in Dafny.

CSE 590: Advanced Programming Languages (UMich)

Winter 2025

- Implemented an automated test-input generator for C programs in OCaml as part of a two-person team.
- Implemented an automated random test-input generation technique based on the DART project.

TEACHING EXPERIENCE

EECS 376: Foundations of Computer Science (UMich)

Graduate Student Instructor

Spring 2025

EECS 270: Introduction to Logic Design (UMich)

Graduate Student Instructor

Fall 2024, Winter 2025

Logic Synthesis and Verification (NTU)

Teaching Assistant

09/2021-01/2022

HONORS AND AWARDS

2021 Research Creativity Award, Ministry of Science and Technology of Taiwan (awarded to 200/3000+ research works)

2020 1st Place, Problem A, 2020 CAD Contest at ICCAD (186 teams competed in 3 problems)

2020 College Student Research Scholarship, Ministry of Science and Technology of Taiwan

SKILLS

Programming Languages C/C++, Python, Go, OCaml, Verilog

Systems Linux, Shell scripting, Git, Docker, debuggers (gdb, pdb, ocamldebug), profilers (valgrind, perf, cProfile), $\text{\LaTeX}$

Verification Dafny, SAT/SMT solvers (Z3, CaDiCaL, Kissat, PySAT), Rocq/Coq, IVy, Berkeley ABC Logic Synthesis and Verification System

Languages Mandarin (Native), English (Proficient, TOEFL: 114/120)